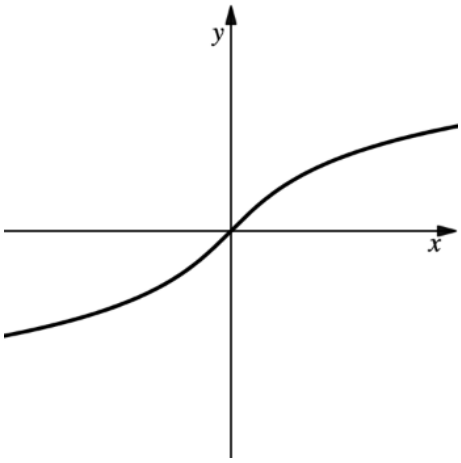
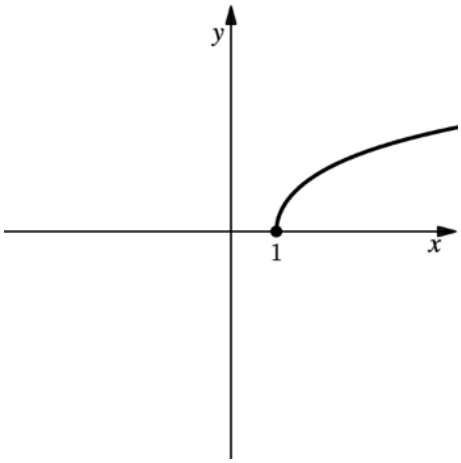
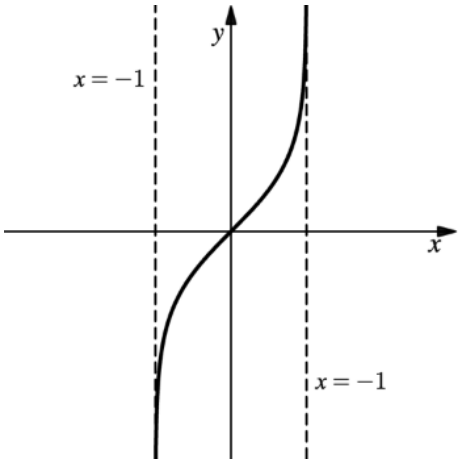
1) a)



1) b)



1) c)



2)

Let $y = \sinh x = \frac{e^x - e^{-x}}{2}$

Swap xs and ys: $x = \frac{e^y - e^{-y}}{2}$

$\Rightarrow x = \frac{e^{2y} - 1}{2e^y}$

$\Rightarrow 2xe^y = e^{2y} - 1$

$\Rightarrow e^{2y} - 2xe^y - 1 = 0$

$\Rightarrow (e^y - x)^2 - x^2 - 1 = 0$

$\Rightarrow (e^y - x)^2 = x^2 + 1$

$\Rightarrow e^y - x = \pm\sqrt{x^2 + 1}$

$\Rightarrow e^y = x \pm \sqrt{x^2 + 1}$

We take the positive square root only, as $e^y > 0$ ($x - \sqrt{x^2 + 1} < 0$ for all real values of x)

$\Rightarrow y = \ln\left(x + \sqrt{x^2 + 1}\right)$

$\therefore \operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$

3)

Let $\frac{y}{k} = \sinh x = \frac{e^x - e^{-x}}{2}$

Swap xs and ys: $\frac{x}{k} = \frac{e^y - e^{-y}}{2}$

$\Rightarrow \frac{x}{k} = \frac{e^{2y} - 1}{2e^y}$

$\Rightarrow \frac{2xe^y}{k} = e^{2y} - 1$

$\Rightarrow e^{2y} - \frac{2xe^y}{k} - 1 = 0$

$\Rightarrow \left(e^y - \frac{x}{k}\right)^2 - \frac{x^2}{k^2} - 1 = 0$

$\Rightarrow \left(e^y - \frac{x}{k}\right)^2 = \frac{x^2}{k^2} + 1$

$\Rightarrow e^y - \frac{x}{k} = \pm\sqrt{\frac{x^2}{k^2} + 1}$

$\Rightarrow e^y = \frac{x}{k} \pm \sqrt{\frac{x^2}{k^2} + 1}$

We take the positive square root only, as $e^y > 0$ ($\frac{x}{k} - \sqrt{\frac{x^2}{k^2} + 1} < 0$ for all real values of x and for $k > 0$)

$\Rightarrow y = \ln\left(\frac{x}{k} + \sqrt{\frac{x^2}{k^2} + 1}\right)$

$\Rightarrow y = \ln\left(\frac{x}{k} + \sqrt{\frac{1}{k^2}(x^2 + k^2)}\right)$

$\Rightarrow y = \ln\left(\frac{x}{k} + \frac{1}{k}\sqrt{x^2 + k^2}\right)$

$\Rightarrow y = \ln\left(\frac{x + \sqrt{x^2 + k^2}}{k}\right)$

$\therefore \operatorname{arsinh}\frac{x}{k} = \ln\left(\frac{x + \sqrt{x^2 + k^2}}{k}\right)$

4)

Let $y = \cosh x = \frac{e^x + e^{-x}}{2}$

Swap xs and ys: $x = \frac{e^y + e^{-y}}{2}$

$\Rightarrow x = \frac{e^{2y}+1}{2e^y}$

$\Rightarrow 2xe^y = e^{2y} + 1$

$\Rightarrow e^{2y} - 2xe^y + 1 = 0$

$\Rightarrow (e^y - x)^2 - x^2 + 1 = 0$

$\Rightarrow (e^y - x)^2 = x^2 - 1$

$\Rightarrow e^y - x = \pm\sqrt{x^2 - 1}$

$\Rightarrow e^y = x \pm \sqrt{x^2 - 1}$

By definition, for $y = \operatorname{arcosh} x$ means that $y \geq 0$

Therefore, $e^y \geq 1$, so the correct root must be at least 1

For $x \geq 1$, $x + \sqrt{x^2 - 1} \geq x \geq 1$, so this works.

For $x > 1$, $\sqrt{x^2 - 1} < x$ and $x - \sqrt{x^2 - 1} < 1$, hence the negative root cannot equal e^y

$\Rightarrow y = \ln\left(x + \sqrt{x^2 - 1}\right)$

$\therefore \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$

5)

Let $\frac{y}{k} = \cosh x = \frac{e^x + e^{-x}}{2}$

Swap xs and ys: $\frac{x}{k} = \frac{e^y + e^{-y}}{2}$

$\Rightarrow \frac{x}{k} = \frac{e^{2y}+1}{2e^y}$

$\Rightarrow \frac{2xe^y}{k} = e^{2y} + 1$

$\Rightarrow e^{2y} - \frac{2xe^y}{k} + 1 = 0$

$\Rightarrow \left(e^y - \frac{x}{k}\right)^2 - \frac{x^2}{k^2} + 1 = 0$

$\Rightarrow \left(e^y - \frac{x}{k}\right)^2 = \frac{x^2}{k^2} - 1$

$\Rightarrow e^y - \frac{x}{k} = \pm\sqrt{\frac{x^2}{k^2} - 1}$

$\Rightarrow e^y = \frac{x}{k} \pm \sqrt{\frac{x^2}{k^2} - 1}$

By definition, for $y = \operatorname{arcosh}\frac{x}{k}$ means that $y \geq 0$

Therefore, $e^y \geq 1$, so the correct root must be at least 1

For $\frac{x}{k} \geq 1$, $\frac{x}{k} + \sqrt{\frac{x^2}{k^2} - 1} \geq \frac{x}{k} \geq 1$, so this works.

For $\frac{x}{k} > 1$, $\sqrt{\frac{x^2}{k^2} - 1} < \frac{x}{k}$ and $\frac{x}{k} - \sqrt{\frac{x^2}{k^2} - 1} < 1$, hence the negative root cannot equal e^y

$\Rightarrow y = \ln\left(\frac{x}{k} + \sqrt{\frac{x^2}{k^2} - 1}\right)$

$\Rightarrow y = \ln\left(\frac{x}{k} + \sqrt{\frac{1}{k^2}(x^2 - k^2)}\right)$

$\Rightarrow y = \ln\left(\frac{x}{k} + \frac{1}{k}\sqrt{x^2 - k^2}\right)$

$\Rightarrow y = \ln\left(\frac{x+\sqrt{x^2-k^2}}{k}\right)$

$\therefore \operatorname{arcosh}\frac{x}{k} = \ln\left(\frac{x+\sqrt{x^2-k^2}}{k}\right)$

6)

Let $y = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Swap xs and ys: $x = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$\Rightarrow x = \frac{e^{2y} - 1}{e^{2y} + 1}$

$\Rightarrow xe^{2y} + x = e^{2y} - 1$

$\Rightarrow 1 + x = e^{2y} - xe^{2y}$

$\Rightarrow 1 + x = e^{2y}(1 - x)$

$\Rightarrow e^{2y} = \frac{1+x}{1-x}$

$\Rightarrow 2y = \ln\left(\frac{1+x}{1-x}\right)$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

$\therefore \operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

7)

Let $\frac{y}{k} = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Swap xs and ys: $\frac{x}{k} = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$\Rightarrow \frac{x}{k} = \frac{e^{2y} - 1}{e^{2y} + 1}$

$\Rightarrow \frac{xe^{2y}}{k} + \frac{x}{k} = e^{2y} - 1$

$\Rightarrow 1 + \frac{x}{k} = e^{2y} - \frac{xe^{2y}}{k}$

$\Rightarrow 1 + \frac{x}{k} = e^{2y}\left(1 - \frac{x}{k}\right)$

$\Rightarrow e^{2y} = \frac{1+\frac{x}{k}}{1-\frac{x}{k}}$

$\Rightarrow e^{2y} = \frac{k+x}{k-x}$

$\Rightarrow 2y = \ln\left(\frac{k+x}{k-x}\right)$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{k+x}{k-x}\right)$

$\therefore \operatorname{artanh}\frac{x}{k} = \frac{1}{2}\ln\left(\frac{k+x}{k-x}\right)$

8)

$\operatorname{arcosh} x = \ln 10$

$\Rightarrow \ln\left(x + \sqrt{x^2 - 1}\right) = \ln 10$

$\Rightarrow x + \sqrt{x^2 - 1} = 10$

$\Rightarrow \sqrt{x^2 - 1} = 10 - x$

$\Rightarrow x^2 - 1 = (10 - x)^2$

$\Rightarrow x^2 - 1 = 100 - 20x + x^2$

$\Rightarrow -1 = 100 - 20x$

$\Rightarrow 20x = 101$

$\Rightarrow x = \frac{101}{20}$

Alternatively:

$\operatorname{arcosh} x = \ln 10$

$\Rightarrow \cosh(\operatorname{arccosh} x) = \cosh(\ln 10)$

$x = \frac{e^{\ln 10} + e^{-\ln 10}}{2}$

$\Rightarrow x = \frac{e^{\ln 10} + e^{\ln \frac{1}{10}}}{2}$

$\Rightarrow x = \frac{10 + \frac{1}{10}}{2}$

$\Rightarrow x = \frac{101}{20}$

9)

$\operatorname{arsinh} x = -\ln 6$

$\Rightarrow \ln\left(x + \sqrt{x^2+1}\right) = \ln 6^{-1} = \ln\frac{1}{6}$

$\Rightarrow x + \sqrt{x^2+1} = \frac{1}{6}$

$\Rightarrow 6x + 6\sqrt{x^2+1} = 1$

$\Rightarrow 6\sqrt{x^2+1} = 1 - 6x$

$\Rightarrow 36(x^2+1) = (1-6x)^2$

$\Rightarrow 36x^2 + 36 = 1 - 12x + 36x^2$

$\Rightarrow 36 = 1 - 12x$

$\Rightarrow 12x = -35$

$\Rightarrow x = -\frac{35}{12}$

Alternatively:

$\operatorname{arsinh} x = -\ln 6$

$\Rightarrow \sinh(\operatorname{arsinh} x) = \sinh(-\ln 6)$

$x = \frac{e^{-\ln 6} - e^{\ln 6}}{2}$

$\Rightarrow x = \frac{e^{\ln\frac{1}{6}} - e^{\ln 6}}{2}$

$\Rightarrow x = \frac{\frac{1}{6} - 6}{2}$

$\Rightarrow x = -\frac{35}{12}$

10)

$\operatorname{artanh}\frac{x}{6} = -\ln 2$

$\Rightarrow \frac{1}{2}\ln\left(\frac{6+x}{6-x}\right) = \ln 2^{-1}$

$\Rightarrow \ln\left(\frac{6+x}{6-x}\right) = 2\ln\frac{1}{2}$

$\Rightarrow \ln\left(\frac{6+x}{6-x}\right) = \ln\left(\frac{1}{2}\right)^2 = \ln\frac{1}{4}$

$\Rightarrow \frac{6+x}{6-x} = \frac{1}{4}$

$\Rightarrow 24 + 4x = 6 - x$

$\Rightarrow 5x = -18$

$\Rightarrow x = -\frac{18}{5}$

Alternatively:

$\operatorname{artanh}\frac{x}{6} = -\ln 2$

$\tanh\left(\operatorname{artanh}\frac{x}{6}\right) = \tanh(-\ln 2)$

$\frac{x}{6} = \frac{e^{-\ln 2} - e^{\ln 2}}{e^{-\ln 2} + e^{\ln 2}}$

$\Rightarrow \frac{x}{6} = \frac{e^{\ln\frac{1}{2}} - e^{\ln 2}}{e^{\ln\frac{1}{2}} + e^{\ln 2}}$

$\Rightarrow \frac{x}{6} = \frac{\frac{1}{2} - 2}{\frac{1}{2} + 2}$

$\Rightarrow \frac{x}{6} = -\frac{3}{5}$

$\Rightarrow x = -\frac{18}{5}$

11) a)

$\operatorname{artanh} x = \ln\left(\sqrt{x+2}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \ln\left(\sqrt{x+2}\right)$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) = 2\ln\left(\sqrt{x+2}\right)$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) = \ln\left(\left(\sqrt{x+2}\right)^2\right)$

$\Rightarrow \frac{1+x}{1-x} = x+2$

$\Rightarrow 1+x = (x+2)(1-x)$

$\Rightarrow 1+x = -x^2 - x + 2$

$\Rightarrow x^2 + 2x - 1 = 0$

$\Rightarrow x = -1 \pm \sqrt{2}$

$x \neq -1-\sqrt{2}$ as $-1 < x < 1$

So $x = -1+\sqrt{2}$ is the only real solution

Alternatively:

$\operatorname{artanh} x = \ln\left(\sqrt{x+2}\right)$

$\tan(\operatorname{artanh} x) = \tan\left(\ln\left(\sqrt{x+2}\right)\right)$

$x = \frac{e^{\ln(\sqrt{x+2})} - e^{-\ln(\sqrt{x+2})}}{e^{\ln(\sqrt{x+2})} + e^{-\ln(\sqrt{x+2})}}$

$\Rightarrow x = \frac{\sqrt{x+2} - \frac{1}{\sqrt{x+2}}}{\sqrt{x+2} + \frac{1}{\sqrt{x+2}}}$

$\Rightarrow x = \frac{x+2-1}{x+2+1}$

$\Rightarrow x = \frac{x+1}{x+3}$

$\Rightarrow x(x+3) = x+1$

$\Rightarrow x^2 + 3x - x - 1 = 0$

$\Rightarrow x^2 + 2x - 1 = 0$

$\Rightarrow x = -1 \pm \sqrt{2}$

$x \neq -1-\sqrt{2}$ as $-1 < x < 1$

So $x = -1+\sqrt{2}$ is the only real solution

11) b)

$\operatorname{artanh}\frac{x}{3} = \ln\left(\sqrt{3-2x}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{3+x}{3-x}\right) = \ln\left(\sqrt{3-2x}\right)$

$\Rightarrow \ln\left(\frac{3+x}{3-x}\right) = 2\ln\left(\sqrt{3-2x}\right)$

$\Rightarrow \ln\left(\frac{3+x}{3-x}\right) = \ln\left(\left(\sqrt{3-2x}\right)^2\right)$

$\Rightarrow \frac{3+x}{3-x} = 3-2x$

$\Rightarrow 3+x = (3-2x)(3-x)$

$\Rightarrow 3+x = 2x^2-9x+9$

$\Rightarrow 0 = 2x^2-10x+6$

$\Rightarrow x = \frac{5}{2} \pm \frac{\sqrt{13}}{2}$

$x \neq \frac{5}{2}+\frac{\sqrt{13}}{2}$ as $\operatorname{artanh}\frac{x}{3}$ is only valid for $-3 < x < 3$

So $x = \frac{5}{2}-\frac{\sqrt{13}}{2}$ is the only real solution

Alternatively:

$\operatorname{artanh}\frac{x}{3} = \ln\left(\sqrt{3-2x}\right)$

$\tan\left(\operatorname{artanh}\frac{x}{3}\right) = \tan\left(\ln\left(\sqrt{3-2x}\right)\right)$

$\frac{x}{3} = \frac{e^{\ln\left(\sqrt{3-2x}\right)}-e^{-\ln\left(\sqrt{3-2x}\right)}}{e^{\ln\left(\sqrt{3-2x}\right)}+e^{-\ln\left(\sqrt{3-2x}\right)}}$

$\Rightarrow \frac{x}{3} = \frac{\sqrt{3-2x}-\frac{1}{\sqrt{3-2x}}}{\sqrt{3-2x}+\frac{1}{\sqrt{3-2x}}}$

$\Rightarrow \frac{x}{3} = \frac{3-2x-1}{3-2x+1}$

$\Rightarrow \frac{x}{3} = \frac{2-2x}{4-2x}$

$\Rightarrow x(4-2x) = 3(2-2x)$

$\Rightarrow 4x-2x^2 = 6-6x$

$\Rightarrow 0 = 2x^2-10x+6$

$\Rightarrow x = \frac{5}{2} \pm \frac{\sqrt{13}}{2}$

$x \neq \frac{5}{2}+\frac{\sqrt{13}}{2}$ as $\operatorname{artanh}\frac{x}{3}$ is only valid for $-3 < x < 3$

So $x = \frac{5}{2}-\frac{\sqrt{13}}{2}$ is the only real solution

11) c)

$\operatorname{artanh} 5x = \ln\left(\sqrt{2+5x}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+5x}{1-5x}\right) = \ln\left(\sqrt{2+5x}\right)$

$\Rightarrow \ln\left(\frac{1+5x}{1-5x}\right) = 2\ln\left(\sqrt{2+5x}\right)$

$\Rightarrow \ln\left(\frac{1+5x}{1-5x}\right) = \ln\left(\left(\sqrt{2+5x}\right)^2\right)$

$\Rightarrow \frac{1+5x}{1-5x} = 2+5x$

$\Rightarrow 1+5x = (2+5x)(1-5x)$

$\Rightarrow 1+5x = -25x^2 - 5x + 2$

$\Rightarrow 25x^2 + 10x - 1 = 0$

$\Rightarrow x = -\frac{1}{5} \pm \frac{\sqrt{2}}{5}$

$x \neq -\frac{1}{5} - \frac{\sqrt{2}}{5}$ as $\operatorname{artanh} 5x$ is only valid for $-\frac{1}{5} < x < \frac{1}{5}$

So $x = -\frac{1}{5} + \frac{\sqrt{2}}{5}$ is the only real solution

Alternatively:

$\operatorname{artanh} 5x = \ln\left(\sqrt{2+5x}\right)$

$\tan(\operatorname{artanh} 5x) = \tan\left(\ln\left(\sqrt{2+5x}\right)\right)$

$5x = \frac{e^{\ln(\sqrt{2+5x})} - e^{-\ln(\sqrt{2+5x})}}{e^{\ln(\sqrt{2+5x})} + e^{-\ln(\sqrt{2+5x})}}$

$\Rightarrow 5x = \frac{\sqrt{2+5x} - \frac{1}{\sqrt{2+5x}}}{\sqrt{2+5x} + \frac{1}{\sqrt{2+5x}}}$

$\Rightarrow 5x = \frac{2+5x-1}{2+5x+1}$

$\Rightarrow 5x = \frac{1+5x}{3+5x}$

$\Rightarrow 5x(3+5x) = 1+5x$

$\Rightarrow 15x + 25x^2 = 1+5x$

$\Rightarrow 25x^2 + 10x - 1 = 0$

$\Rightarrow x = -\frac{1}{5} \pm \frac{\sqrt{2}}{5}$

$x \neq -\frac{1}{5} - \frac{\sqrt{2}}{5}$ as $\operatorname{artanh} 5x$ is only valid for $-\frac{1}{5} < x < \frac{1}{5}$

So $x = -\frac{1}{5} + \frac{\sqrt{2}}{5}$ is the only real solution

11) d)

$4x+1=\tanh\left(\ln\left(\sqrt{5-2x}\right)\right)$

$\Rightarrow \operatorname{artanh}(4x+1)=\ln\left(\sqrt{5-2x}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+(4x+1)}{1-(4x+1)}\right)=\ln\left(\sqrt{5-2x}\right)$

$\Rightarrow \ln\left(\frac{2+4x}{-4x}\right)=2\ln\left(\sqrt{5-2x}\right)$

$\Rightarrow \ln\left(\frac{2+4x}{-4x}\right)=\ln\left(\left(\sqrt{5-2x}\right)^2\right)$

$\Rightarrow \frac{2+4x}{-4x}=5-2x$

$\Rightarrow 2+4x=(5-2x)(-4x)$

$\Rightarrow 2+4x=8x^2-20x$

$\Rightarrow 0=8x^2-24x-2$

$\Rightarrow x=\frac{3}{2}\pm\frac{\sqrt{10}}{2}$

$x\neq\frac{3}{2}+\frac{\sqrt{10}}{2}$ as $\operatorname{artanh}(4x+1)$ is only valid for $-1<4x+1<1\Rightarrow-\frac{1}{2}<x<0$

So $x=\frac{3}{2}-\frac{\sqrt{10}}{2}$ is the only real solution

Alternatively:

$4x+1=\tanh\left(\ln\left(\sqrt{5-2x}\right)\right)$

$4x+1=\frac{e^{\ln\left(\sqrt{5-2x}\right)}-e^{-\ln\left(\sqrt{5-2x}\right)}}{e^{\ln\left(\sqrt{5-2x}\right)}+e^{-\ln\left(\sqrt{5-2x}\right)}}$

$\Rightarrow 4x+1=\frac{\sqrt{5-2x}-\frac{1}{\sqrt{5-2x}}}{\sqrt{5-2x}+\frac{1}{\sqrt{5-2x}}}$

$\Rightarrow 4x+1=\frac{5-2x-1}{5-2x+1}$

$\Rightarrow 4x+1=\frac{4-2x}{6-2x}$

$\Rightarrow (4x+1)(6-2x)=4-2x$

$\Rightarrow 24x-8x^2+6-2x=4-2x$

$\Rightarrow 0=8x^2-24x-2$

$\Rightarrow x=\frac{3}{2}\pm\frac{\sqrt{10}}{2}$

$x\neq\frac{3}{2}+\frac{\sqrt{10}}{2}$ as $\operatorname{artanh}(4x+1)$ is only valid for $-1<4x+1<1\Rightarrow-\frac{1}{2}<x<0$

So $x=\frac{3}{2}-\frac{\sqrt{10}}{2}$ is the only real solution

12) a)

$y = \operatorname{arsinh} x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}\}$

12) b)

$y = \operatorname{arcosh} x$

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

12) c)

$y = \operatorname{artanh} x$

Domain: $\{x: x \in \mathbb{R}, -1 < x < 1\}$

Range: $\{y: y \in \mathbb{R}\}$

13) a)

$$\mathrm{f}\left(-\tfrac{1}{2}\right) = \ln\left(-\tfrac{1}{2} + \sqrt{\left(-\tfrac{1}{2}\right)^2 + 1}\right) = \ln\left(-\tfrac{1}{2} + \tfrac{\sqrt{5}}{2}\right)$$

$$\mathrm{f}\left(\tfrac{1}{2}\right) = \ln\left(\tfrac{1}{2} + \sqrt{\left(\tfrac{1}{2}\right)^2 + 1}\right) = \ln\left(\tfrac{1}{2} + \tfrac{\sqrt{5}}{2}\right)$$

Range: $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \ln\left(-\tfrac{1}{2} + \tfrac{\sqrt{5}}{2}\right) < \mathrm{f}(x) < \ln\left(\tfrac{1}{2} + \tfrac{\sqrt{5}}{2}\right)\right\}$

13) b)

$$\mathrm{g}\left(-\tfrac{1}{2}\right) = \tfrac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \tfrac{1}{2}\ln\left(\frac{\frac{1}{2}}{\frac{3}{2}}\right) = \tfrac{1}{2}\ln\tfrac{1}{3} = -\ln\sqrt{3}$$

$$\mathrm{g}\left(\tfrac{1}{2}\right) = \tfrac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \tfrac{1}{2}\ln\left(\frac{\frac{3}{2}}{\frac{1}{2}}\right) = \tfrac{1}{2}\ln 3 = \ln\sqrt{3}$$

Range: $\left\{\mathrm{g}(x): \mathrm{g}(x) \in \mathbb{R}, -\ln\sqrt{3} < \mathrm{g}(x) < \ln\sqrt{3}\right\}$

14) a)

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 1\}$

14) b)

Domain: $\{x: x \in \mathbb{R}, x \geq 2\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) c)

Domain: $\left\{x: x \in \mathbb{R}, x \geq \frac{1}{2}\right\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) d)

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) e)

Domain: $\{x: x \in \mathbb{R}, x \geq -1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) f)

Domain: $\{x: x \in \mathbb{R}, x \geq 2\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) g)

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq -5\}$

14) h)

Domain: $\{x: x \in \mathbb{R}, x \leq 3\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

14) i)

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \leq 8\}$

14) j)

Domain: $\{x: x \in \mathbb{R}, x \leq 16\}$

Range: $\{y: y \in \mathbb{R}, y \leq 0\}$

15) a)

$y = \text{arsinh}^2 x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

15) b)

$y = \text{arcosh}^2 x$

Domain: $\{x: x \in \mathbb{R}, x \geq 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

15) c)

$y = \text{artanh}^2 x$

Domain: $\{x: x \in \mathbb{R}, -1 < x < 1\}$

Range: $\{y: y \in \mathbb{R}, y \geq 0\}$

16)

$\text{arsinh}\, x = \ln\left(x + \sqrt{x^2 + 1}\right)$ and $\text{arcosh}\, x = \ln\left(x + \sqrt{x^2 - 1}\right)$

Since $\sqrt{x^2 + 1} > \sqrt{x^2 - 1}$ for all $x \geq 1$, it must be that $\text{arsinh}\, x > \text{arcosh}\, x$ for all $x \geq 1$

17)

$\text{f}(1) = \text{arsinh}\, 1 + \text{arcosh}\, 1$

$= \ln\left(1 + \sqrt{1^2 + 1}\right) + \ln\left(1 + \sqrt{1^2 - 1}\right)$

$= \ln\left(1 + \sqrt{2}\right) + \ln 1$

$= \ln\left(1 + \sqrt{2}\right)$

$\therefore$ Range: $\left\{\text{f}(x): \text{f}(x) \in \mathbb{R}, \text{f}(x) \geq \ln\left(1 + \sqrt{2}\right)\right\}$

18)

$\operatorname{artanh} x + \ln\left(\sqrt{1-x}\right) = \frac{1}{2}\ln 3 - 2\ln 2$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) + \ln(1-x)^{\frac{1}{2}} = \frac{1}{2}\ln 3 - 2\ln 2$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} + \ln(1-x)^{\frac{1}{2}} = \ln 3^{\frac{1}{2}} - \ln 2^2$

$\Rightarrow \ln\left(\frac{(1+x)^{\frac{1}{2}}}{(1-x)^{\frac{1}{2}}} \times (1-x)^{\frac{1}{2}}\right) = \ln\left(\frac{3^{\frac{1}{2}}}{4}\right)$

$\Rightarrow (1+x)^{\frac{1}{2}} = \frac{3^{\frac{1}{2}}}{4}$

$\Rightarrow 1 + x = \frac{3}{16}$

$\Rightarrow x = -\frac{13}{16}$

19)

$\operatorname{arsinh} x = \ln 4 - \ln 3 + \operatorname{arcosh} x$

$\Rightarrow \operatorname{arsinh} x - \operatorname{arcosh} x = \ln\frac{4}{3}$

$\Rightarrow \ln\left(x + \sqrt{x^2+1}\right) - \ln\left(x + \sqrt{x^2-1}\right) = \ln\frac{4}{3}$

$\Rightarrow \ln\left(\frac{x+\sqrt{x^2+1}}{x+\sqrt{x^2-1}}\right) = \ln\frac{4}{3}$

$\Rightarrow \frac{x+\sqrt{x^2+1}}{x+\sqrt{x^2-1}} = \frac{4}{3}$

$\Rightarrow 3x + 3\sqrt{x^2+1} = 4x + 4\sqrt{x^2-1}$

$\Rightarrow 3\sqrt{x^2+1} - 4\sqrt{x^2-1} = x$

$\Rightarrow \left(3\sqrt{x^2+1} - 4\sqrt{x^2-1}\right)^2 = x^2$

$\Rightarrow 9(x^2+1) - 24\sqrt{x^4-1} + 16(x^2-1) = x^2$

$\Rightarrow 24x^2 - 7 = 24\sqrt{x^4-1}$

$\Rightarrow (24x^2-7)^2 = \left(24\sqrt{x^4-1}\right)^2$

$\Rightarrow 576x^4 - 336x^2 + 49 = 576(x^4-1)$

$\Rightarrow -336x^2 + 625 = 0$

$\Rightarrow x^2 = \frac{625}{336}$

$\Rightarrow x = \frac{25}{4\sqrt{21}} = \frac{25\sqrt{21}}{84}$

20)

$\mathrm{f}(x) = \sqrt{x^2+1} + \operatorname{arsinh} x$

$= (x^2+1)^{\frac{1}{2}} + \operatorname{arsinh} x$

$\mathrm{f}'(x) = \frac{1}{2} \times 2x \times (x^2+1)^{-\frac{1}{2}} + \frac{1}{\sqrt{x^2+1}}$

$= \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}$

$= \frac{x+1}{\sqrt{x^2+1}}$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow \frac{x+1}{\sqrt{x^2+1}} = 0 \Rightarrow x+1 = 0 \Rightarrow x = -1$

$\mathrm{f}(-1) = \sqrt{(-1)^2+1} + \ln\left(-1 + \sqrt{(-1)^2+1}\right)$

$= \sqrt{2} + \ln\left(-1+\sqrt{2}\right)$

Range is $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \sqrt{2} + \ln\left(-1+\sqrt{2}\right)\right\}$

21)

$\mathrm{f}(x) = \sqrt{x^2-1} - 4\operatorname{arcosh} x$

$= (x^2-1)^{\frac{1}{2}} - 4\operatorname{arcosh} x$

$\mathrm{f}'(x) = \frac{1}{2} \times 2x \times (x^2-1)^{-\frac{1}{2}} - \frac{4}{\sqrt{x^2-1}}$

$= \frac{x}{\sqrt{x^2-1}} - \frac{4}{\sqrt{x^2-1}}$

$= \frac{x-4}{\sqrt{x^2-1}}$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow \frac{x-4}{\sqrt{x^2-1}} = 0 \Rightarrow x-4 = 0 \Rightarrow x = 4$

$\mathrm{f}(4) = \sqrt{4^2-1} - 4\ln\left(4+\sqrt{4^2-1}\right)$

$= \sqrt{15} - 4\ln\left(4+\sqrt{15}\right)$

Range is $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \sqrt{15} - 4\ln\left(4+\sqrt{15}\right)\right\}$

22) a)

The largest possible domain of $\operatorname{arsinh} x$ is $\{x: x \in \mathbb{R}\}$, while the largest possible domain of $\operatorname{arcosh} x$ is $\{x: x \in \mathbb{R}, x \geq 1\}$. So the largest possible domain of $\mathrm{f}(x) = 2\operatorname{arsinh} x - \operatorname{arcosh} x$ must be $\{x: x \in \mathbb{R}, x \geq 1\}$ as this satisfies **both** of the inverse hyperbolic terms in the function.

22) b)

$\mathrm{f}(x) = 2\,\mathrm{arsinh}\,x - \mathrm{arcosh}\,x$

$\mathrm{f}'(x) = 2 \times \frac{1}{\sqrt{x^2+1}} - \frac{1}{\sqrt{x^2-1}}$

$= \frac{2}{\sqrt{x^2+1}} - \frac{1}{\sqrt{x^2-1}}$

$= \frac{2\sqrt{x^2-1}-\sqrt{x^2+1}}{\sqrt{x^2+1}\sqrt{x^2-1}}$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow \frac{2\sqrt{x^2-1}-\sqrt{x^2+1}}{\sqrt{x^2+1}\sqrt{x^2-1}} = 0$

$\Rightarrow 2\sqrt{x^2-1} - \sqrt{x^2+1} = 0$

$\Rightarrow 2\sqrt{x^2-1} = \sqrt{x^2+1}$

$\Rightarrow 4(x^2-1) = x^2+1$

$\Rightarrow 4x^2 - 4 = x^2 + 1$

$\Rightarrow 3x^2 = 5$

$\Rightarrow x^2 = \frac{5}{3}$

$\Rightarrow x = \pm\sqrt{\frac{5}{3}}$

As $x \geq 1$, there is only one stationary point at $x = \sqrt{\frac{5}{3}}$

22) c)

$$\mathrm{f}\left(\sqrt{\frac{5}{3}}\right) = 2\ln\left(\sqrt{\frac{5}{3}} + \sqrt{\left(\sqrt{\frac{5}{3}}\right)^2 + 1}\right) - \ln\left(\sqrt{\frac{5}{3}} + \sqrt{\left(\sqrt{\frac{5}{3}}\right)^2 - 1}\right)$$

$$= 2\ln\left(\sqrt{\frac{5}{3}} + \frac{2\sqrt{6}}{3}\right) - \ln\left(\sqrt{\frac{5}{3}} + \frac{\sqrt{6}}{3}\right)$$

$$= 2\ln\left(\frac{\sqrt{15}}{3} + \frac{2\sqrt{6}}{3}\right) - \ln\left(\frac{\sqrt{15}}{3} + \frac{\sqrt{6}}{3}\right)$$

$$= 2\ln\left(\frac{\sqrt{15}+2\sqrt{6}}{3}\right) - \ln\left(\frac{\sqrt{15}+\sqrt{6}}{3}\right)$$

$$= 2\left(\ln\left(\sqrt{15}+2\sqrt{6}\right) - \ln 3\right) - \left(\ln\left(\sqrt{15}+\sqrt{6}\right) - \ln 3\right)$$

$$= 2\ln\left(\sqrt{15}+2\sqrt{6}\right) - \ln\left(\sqrt{15}+\sqrt{6}\right) - \ln 3$$

Range is $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 2\ln\left(\sqrt{15}+2\sqrt{6}\right) - \ln\left(\sqrt{15}+\sqrt{6}\right) - \ln 3\right\}$

23)

Let $y = \text{cosech}\, x = \frac{2}{e^x - e^{-x}}$

Swap xs and ys: $x = \frac{2}{e^y - e^{-y}}$

$\Rightarrow x = \frac{2e^y}{e^{2y} - 1}$

$\Rightarrow xe^{2y} - x = 2e^y$

$\Rightarrow xe^{2y} - 2e^y - x = 0$

$\Rightarrow e^{2y} - \frac{2}{x}e^y - 1 = 0$

$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 - \frac{1}{x^2} - 1 = 0$

$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 = \frac{1}{x^2} + 1$

$\Rightarrow e^y - \frac{1}{x} = \pm\sqrt{\frac{1}{x^2} + 1}$

$\Rightarrow e^y = \frac{1}{x} \pm \sqrt{\frac{1}{x^2} + 1}$

We take the positive square root only, as $e^y > 0$ ($\frac{1}{x} - \sqrt{\frac{1}{x^2} + 1} < 0$ for all real values of $x \neq 0$)

$\Rightarrow y = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right)$

$\therefore \text{arcosech}\, x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right)$

24)

Let $\frac{y}{k} = \text{cosech}\, x = \frac{2}{e^x - e^{-x}}$

Swap xs and ys: $\frac{x}{k} = \frac{2}{e^y - e^{-y}}$

$\Rightarrow \frac{x}{k} = \frac{2e^y}{e^{2y} - 1}$

$\Rightarrow \frac{xe^{2y}}{k} - \frac{x}{k} = 2e^y$

$\Rightarrow \frac{xe^{2y}}{k} - 2e^y - \frac{x}{k} = 0$

$\Rightarrow e^{2y} - \frac{2k}{x}e^y - 1 = 0$

$\Rightarrow \left(e^y - \frac{k}{x}\right)^2 - \frac{k^2}{x^2} - 1 = 0$

$\Rightarrow \left(e^y - \frac{k}{x}\right)^2 = \frac{k^2}{x^2} + 1$

$\Rightarrow e^y - \frac{k}{x} = \pm\sqrt{\frac{k^2}{x^2} + 1}$

$\Rightarrow e^y = \frac{k}{x} \pm \sqrt{\frac{k^2}{x^2} + 1}$

We take the positive square root only, as $e^y > 0$ ($\frac{k}{x} - \sqrt{\frac{k^2}{x^2} + 1} < 0$ for all real values of $x \neq 0$ and for $k > 0$)

$\Rightarrow y = \ln\left(\frac{k}{x} + \sqrt{\frac{k^2}{x^2} + 1}\right)$

$\Rightarrow y = \ln\left(\frac{k}{x} + \sqrt{\frac{1}{x^2}(k^2 + x^2)}\right)$

$\Rightarrow y = \ln\left(\frac{k}{x} + \frac{1}{x}\sqrt{k^2 + x^2}\right)$

$\Rightarrow y = \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right)$

$\therefore \text{arcosech}\, \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right)$

25)

Let $y = \operatorname{sech} x = \frac{2}{e^x + e^{-x}}$

Swap xs and ys: $x = \frac{2}{e^y + e^{-y}}$

$\Rightarrow x = \frac{2e^y}{e^{2y}+1}$

$\Rightarrow xe^{2y} + x = 2e^y$

$\Rightarrow xe^{2y} - 2e^y + x = 0$

$\Rightarrow e^{2y} - \frac{2}{x}e^y + 1 = 0$

$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 - \frac{1}{x^2} + 1 = 0$

$\Rightarrow \left(e^y - \frac{1}{x}\right)^2 = \frac{1}{x^2} - 1$

$\Rightarrow e^y - \frac{1}{x} = \pm\sqrt{\frac{1}{x^2} - 1}$

$\Rightarrow e^y = \frac{1}{x} \pm \sqrt{\frac{1}{x^2} - 1}$

Now, $0 < x < 1$ and for $y = \operatorname{arsech} x$ the range is $y \geq 0$

Therefore, $e^y \geq 1$, so the correct root must be at least 1

For $0 < x < 1$, $\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1} \geq \frac{1}{x} > 1$, so this works.

For $0 < x < 1$, $\sqrt{\frac{1}{x^2} - 1} < \frac{1}{x}$ because $\frac{1}{x^2} - 1 < \frac{1}{x^2}$ and so $0 < \frac{1}{x} - \sqrt{\frac{1}{x^2} - 1} < 1$, hence the negative root cannot equal e^y

$\Rightarrow y = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right)$

$\therefore \operatorname{arsech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right)$

26)

Let $\frac{y}{k} = \operatorname{sech} x = \frac{2}{e^{x}+e^{-x}}$

Swap xs and ys: $\frac{x}{k} = \frac{2}{e^{y}+e^{-y}}$

$\Rightarrow \frac{x}{k} = \frac{2e^{y}}{e^{2y}+1}$

$\Rightarrow \frac{xe^{2y}}{k} + \frac{x}{k} = 2e^{y}$

$\Rightarrow \frac{xe^{2y}}{k} - 2e^{y} + \frac{x}{k} = 0$

$\Rightarrow e^{2y} - \frac{2k}{x}e^{y} + 1 = 0$

$\Rightarrow \left(e^{y} - \frac{k}{x}\right)^{2} - \frac{k^{2}}{x^{2}} + 1 = 0$

$\Rightarrow \left(e^{y} - \frac{k}{x}\right)^{2} = \frac{k^{2}}{x^{2}} - 1$

$\Rightarrow e^{y} - \frac{k}{x} = \pm\sqrt{\frac{k^{2}}{x^{2}} - 1}$

$\Rightarrow e^{y} = \frac{k}{x} \pm \sqrt{\frac{k^{2}}{x^{2}} - 1}$

Now, $0 < \frac{x}{k} < 1$ and for $y = \operatorname{arsech}\frac{x}{k}$ the range is $y \geq 0$

Therefore, $e^{y} \geq 1$, so the correct root must be at least 1

For $0 < \frac{x}{k} < 1$, we have $\frac{k}{x} > 1$, so $\frac{k}{x} + \sqrt{\frac{k^{2}}{x^{2}} - 1} \geq \frac{k}{x} > 1$, so this works.

For $0 < \frac{x}{k} < 1$, $\sqrt{\frac{k^{2}}{x^{2}} - 1} < \frac{k}{x}$ because $\frac{k^{2}}{x^{2}} - 1 < \frac{k^{2}}{x^{2}}$ and so $0 < \frac{k}{x} - \sqrt{\frac{k^{2}}{x^{2}} - 1} < 1$, hence the negative root cannot equal e^{y}

$\Rightarrow y = \ln\left(\frac{k}{x} + \sqrt{\frac{k^{2}}{x^{2}} - 1}\right)$

$\Rightarrow y = \ln\left(\frac{k}{x} + \sqrt{\frac{1}{x^{2}}(k^{2} - x^{2})}\right)$

$\Rightarrow y = \ln\left(\frac{k}{x} + \frac{1}{x}\sqrt{k^{2} - x^{2}}\right)$

$\Rightarrow y = \ln\left(\frac{k+\sqrt{k^{2}-x^{2}}}{x}\right)$

$\therefore \operatorname{arsech}\frac{x}{k} = \ln\left(\frac{k+\sqrt{k^{2}-x^{2}}}{x}\right)$

27)

Let $y = \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Swap xs and ys: $x = \frac{e^y + e^{-y}}{e^y - e^{-y}}$

$\Rightarrow x = \frac{e^{2y}+1}{e^{2y}-1}$

$\Rightarrow xe^{2y} - x = e^{2y} + 1$

$\Rightarrow x + 1 = xe^{2y} - e^{2y}$

$\Rightarrow x + 1 = e^{2y}(x - 1)$

$\Rightarrow e^{2y} = \frac{x+1}{x-1}$

$\Rightarrow 2y = \ln\left(\frac{x+1}{x-1}\right)$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$

$\therefore \operatorname{arcoth} x = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$

28)

Let $\frac{y}{k} = \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Swap xs and ys: $\frac{x}{k} = \frac{e^y + e^{-y}}{e^y - e^{-y}}$

$\Rightarrow \frac{x}{k} = \frac{e^{2y}+1}{e^{2y}-1}$

$\Rightarrow \frac{xe^{2y}}{k} - \frac{x}{k} = e^{2y} + 1$

$\Rightarrow \frac{x}{k} + 1 = \frac{xe^{2y}}{k} - e^{2y}$

$\Rightarrow \frac{x}{k} + 1 = e^{2y}\left(\frac{x}{k} - 1\right)$

$\Rightarrow e^{2y} = \frac{\frac{x}{k}+1}{\frac{x}{k}-1}$

$\Rightarrow 2y = \ln\left(\frac{x+k}{x-k}\right)$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{x+k}{x-k}\right)$

$\therefore \operatorname{arcoth}\frac{x}{k} = \frac{1}{2}\ln\left(\frac{x+k}{x-k}\right)$

29)

$\operatorname{arcosech} x = \ln 3$

$\Rightarrow \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right) = \ln 3$

$\Rightarrow \frac{1}{x} + \sqrt{\frac{1}{x^2} + 1} = 3$

$\Rightarrow \frac{1}{x} + \sqrt{\frac{1}{x^2}(1 + x^2)} = 3$

$\Rightarrow \frac{1}{x} + \frac{1}{x}\sqrt{1 + x^2} = 3$

$\Rightarrow 1 + \sqrt{1 + x^2} = 3x$

$\Rightarrow \sqrt{1 + x^2} = 3x - 1$

$\Rightarrow 1 + x^2 = (3x - 1)^2$

$\Rightarrow 1 + x^2 = 9x^2 - 6x + 1$

$\Rightarrow 0 = 8x^2 - 6x$

$\Rightarrow 0 = 4x^2 - 3x$

$\Rightarrow 0 = x(4x - 3)$

$\Rightarrow x = 0$ or $x = \frac{3}{4}$

$x \neq 0$ as arcosech 0 is undefined.

So the only solution is $x = \frac{3}{4}$

Alternatively:

$\operatorname{arcosech} x = \ln 3$

$\operatorname{cosech}(\operatorname{arcosech} x) = \operatorname{cosech}(\ln 3)$

$x = \frac{2}{e^{\ln 3} - e^{-\ln 3}}$

$\Rightarrow x = \frac{2}{e^{\ln 3} - e^{\ln \frac{1}{3}}}$

$\Rightarrow x = \frac{2}{3 - \frac{1}{3}}$

$\therefore x = \frac{3}{4}$

30)

$\operatorname{arsech}\frac{x}{4} = \ln 5$

$\Rightarrow \ln\left(\frac{4+\sqrt{16-x^2}}{x}\right) = \ln 5$

$\Rightarrow \frac{4+\sqrt{16-x^2}}{x} = 5$

$\Rightarrow 4 + \sqrt{16-x^2} = 5x$

$\Rightarrow \sqrt{16-x^2} = 5x - 4$

$\Rightarrow 16 - x^2 = (5x-4)^2$

$\Rightarrow 16 - x^2 = 25x^2 - 40x + 16$

$\Rightarrow 0 = 26x^2 - 40x$

$\Rightarrow 0 = 13x^2 - 20x$

$\Rightarrow 0 = x(13x - 20)$

$\Rightarrow x = 0$ or $x = \frac{20}{13}$

$x \neq 0$ as $\operatorname{arsech} 0$ is undefined.

So the only solution is $x = \frac{20}{13}$

Alternatively:

$\operatorname{arsech}\frac{x}{4} = \ln 5$

$\operatorname{sech}\left(\operatorname{arsech}\frac{x}{4}\right) = \operatorname{sech}(\ln 5)$

$\frac{x}{4} = \frac{2}{e^{\ln 5}+e^{-\ln 5}}$

$\Rightarrow \frac{x}{4} = \frac{2}{e^{\ln 5}+e^{\ln\frac{1}{5}}}$

$\Rightarrow \frac{x}{4} = \frac{2}{5+\frac{1}{5}}$

$\Rightarrow \frac{x}{4} = \frac{5}{13}$

$\therefore x = \frac{20}{13}$

31)

$\operatorname{arcoth} x = \ln 2$

$\Rightarrow \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right) = \ln 2$

$\Rightarrow \ln\left(\frac{x+1}{x-1}\right) = 2\ln 2$

$\Rightarrow \ln\left(\frac{x+1}{x-1}\right) = \ln 4$

$\Rightarrow \frac{x+1}{x-1} = 4$

$\Rightarrow x + 1 = 4x - 4$

$\Rightarrow 3x = 5$

$\Rightarrow x = \frac{5}{3}$

Alternatively:

$\operatorname{arcoth} x = \ln 2$

$\coth(\operatorname{arcoth} x) = \coth(\ln 2)$

$x = \frac{e^{\ln 2} + e^{-\ln 2}}{e^{\ln 2} - e^{-\ln 2}}$

$\Rightarrow x = \frac{e^{\ln 2} + e^{\ln \frac{1}{2}}}{e^{\ln 2} - e^{\ln \frac{1}{2}}}$

$\Rightarrow x = \frac{2 + \frac{1}{2}}{2 - \frac{1}{2}}$

$\therefore x = \frac{5}{3}$

32) a)

$\text{arcosech}\, x = \ln\left(2 + \frac{1}{x}\right)$

$\Rightarrow \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right) = \ln\left(2 + \frac{1}{x}\right)$

$\Rightarrow \frac{1}{x} + \sqrt{\frac{1}{x^2} + 1} = 2 + \frac{1}{x}$

$\Rightarrow \sqrt{\frac{1}{x^2} + 1} = 2$

$\Rightarrow \frac{1}{x^2} + 1 = 4$

$\Rightarrow \frac{1}{x^2} = 3$

$\Rightarrow x^2 = \frac{1}{3}$

$\Rightarrow x = \pm\frac{\sqrt{3}}{3}$

Alternatively:

$\text{arcosech}\, x = \ln\left(2 + \frac{1}{x}\right)$

$\text{cosech}(\text{arcosech}\, x) = \text{cosech}\left(\ln\left(2 + \frac{1}{x}\right)\right)$

$x = \frac{2}{e^{\ln\left(2+\frac{1}{x}\right)} - e^{-\ln\left(2+\frac{1}{x}\right)}}$

$\Rightarrow x = \frac{2}{2+\frac{1}{x} - \frac{1}{2+\frac{1}{x}}}$

$\Rightarrow x = \frac{2}{2+\frac{1}{x} - \frac{x}{2x+1}}$

$\Rightarrow x\left(2 + \frac{1}{x} - \frac{x}{2x+1}\right) = 2$

$\Rightarrow 2x + 1 - \frac{x^2}{2x+1} = 2$

$\Rightarrow (2x+1)^2 - x^2 - 2(2x+1) = 0$

$\Rightarrow 4x^2 + 4x + 1 - x^2 - 4x - 2 = 0$

$\Rightarrow 3x^2 = 1$

$\Rightarrow x^2 = \frac{1}{3}$

$\Rightarrow x = \pm\frac{1}{\sqrt{3}}$

$\therefore x = \pm\frac{\sqrt{3}}{3}$

32) b)

$\text{arsech}\,\frac{x}{2} = \ln\left(3 - \frac{4}{x}\right)$

$\Rightarrow \ln\left(\frac{2+\sqrt{4-x^2}}{x}\right) = \ln\left(3 - \frac{4}{x}\right)$

$\Rightarrow \frac{2+\sqrt{4-x^2}}{x} = 3 - \frac{4}{x}$

$\Rightarrow 2 + \sqrt{4-x^2} = 3x - 4$

$\Rightarrow \sqrt{4-x^2} = 3x - 6$

$\Rightarrow 4 - x^2 = (3x-6)^2$

$\Rightarrow 4 - x^2 = 9x^2 - 36x + 36$

$\Rightarrow 0 = 10x^2 - 36x + 32$

$\Rightarrow 0 = 5x^2 - 18x + 16$

$\Rightarrow x = \frac{8}{5}$ or $x = 2$

$x \neq \frac{8}{5}$ as $\ln\left(\frac{2+\sqrt{4-\left(\frac{8}{5}\right)^2}}{\frac{8}{5}}\right) = \ln 2$ but $\ln\left(3 - \frac{4}{\left(\frac{8}{5}\right)}\right) = \ln\frac{1}{2}$ and $\ln 2 \neq \ln\frac{1}{2}$

So the only real solution is $x = 2$

Alternatively:

$\text{arsech}\,\frac{x}{2} = \ln\left(3 - \frac{4}{x}\right)$

$\text{sech}\left(\text{arsech}\,\frac{x}{2}\right) = \text{sech}\left(\ln\left(3 - \frac{4}{x}\right)\right)$

$\frac{x}{2} = \frac{2}{e^{\ln\left(3-\frac{4}{x}\right)} + e^{-\ln\left(3-\frac{4}{x}\right)}}$

$\Rightarrow \frac{x}{2} = \frac{2}{3-\frac{4}{x}+\frac{1}{3-\frac{4}{x}}}$

$\Rightarrow x = \frac{4}{3-\frac{4}{x}+\frac{x}{3x-4}}$

$\Rightarrow x\left(3 - \frac{4}{x} + \frac{x}{3x-4}\right) = 4$

$\Rightarrow 3x - 4 + \frac{x^2}{3x-4} = 4$

$\Rightarrow (3x-4)^2 + x^2 - 4(3x-4) = 0$

$\Rightarrow 9x^2 - 24x + 16 + x^2 - 12x + 16 = 0$

$\Rightarrow 10x^2 - 36x + 32 = 0$

$\Rightarrow 0 = 5x^2 - 18x + 16$

$\Rightarrow x = \frac{8}{5}$ or $x = 2$

$x \neq \frac{8}{5}$ as $\ln\left(\frac{2+\sqrt{4-\left(\frac{8}{5}\right)^2}}{\frac{8}{5}}\right) = \ln 2$ but $\ln\left(3-\frac{4}{\left(\frac{8}{5}\right)}\right) = \ln\frac{1}{2}$ and $\ln 2 \neq \ln\frac{1}{2}$

So the only real solution is $x = 2$

32) c)

$\operatorname{arcoth} 3x = \ln\left(\sqrt{3+4x}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{3x+1}{3x-1}\right) = \ln\left(\sqrt{3+4x}\right)$

$\Rightarrow \ln\left(\frac{3x+1}{3x-1}\right) = 2\ln\left(\sqrt{3+4x}\right)$

$\Rightarrow \ln\left(\frac{3x+1}{3x-1}\right) = \ln\left(\sqrt{3+4x}\right)^2$

$\Rightarrow \frac{3x+1}{3x-1} = 3+4x$

$\Rightarrow 3x+1 = (3+4x)(3x-1)$

$\Rightarrow 3x+1 = 12x^2+5x-3$

$\Rightarrow 0 = 12x^2+2x-4$

$\Rightarrow 0 = 6x^2+x-2$

$\Rightarrow x = -\frac{2}{3}$ or $x = \frac{1}{2}$

Alternatively:

$\operatorname{arcoth} 3x = \ln\left(\sqrt{3+4x}\right)$

$\coth(\operatorname{arcoth} 3x) = \coth\left(\ln\left(\sqrt{3+4x}\right)\right)$

$3x = \frac{e^{\ln\left(\sqrt{3+4x}\right)}+e^{-\ln\left(\sqrt{3+4x}\right)}}{e^{\ln\left(\sqrt{3+4x}\right)}-e^{-\ln\left(\sqrt{3+4x}\right)}}$

$\Rightarrow 3x = \frac{\sqrt{3+4x}+\frac{1}{\sqrt{3+4x}}}{\sqrt{3+4x}-\frac{1}{\sqrt{3+4x}}}$

$\Rightarrow 3x = \frac{3+4x+1}{3+4x-1}$

$\Rightarrow 3x = \frac{4+4x}{2+4x}$

$\Rightarrow 3x(2+4x) = 4+4x$

$\Rightarrow 6x+12x^2-4-4x = 0$

$\Rightarrow 12x^2+2x-4 = 0$

$\Rightarrow 6x^2+x-2 = 0$

$\Rightarrow x = -\frac{2}{3}$ or $x = \frac{1}{2}$

33)

$\text{artanh}\,\frac{x}{2} = \text{arcoth}(3x+1)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{2+x}{2-x}\right) = \frac{1}{2}\ln\left(\frac{(3x+1)+1}{(3x+1)-1}\right)$

$\Rightarrow \ln\left(\frac{2+x}{2-x}\right) = \ln\left(\frac{3x+2}{3x}\right)$

$\Rightarrow \frac{2+x}{2-x} = \frac{3x+2}{3x}$

$\Rightarrow 3x(2+x) = (3x+2)(2-x)$

$\Rightarrow 6x + 3x^2 = -3x^2 + 4x + 4$

$\Rightarrow 6x^2 + 2x - 4 = 0$

$\Rightarrow 3x^2 + x - 2 = 0$

$\Rightarrow x = -1 \text{ or } x = \frac{2}{3}$